

Youssef Naggar

AI Engineer

LinkedIn • Portfolio • GitHub

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Cairo, Egypt

Education

Cairo University, Faculty of Computers and Artificial Intelligence | B.Sc. in Computer Science - AI Specialization | Expected 2027
Currently a 4th-year student, GPA: 3.58/4.0, Excellent, Top 10 Cohort

Technical Skills

Languages: Python, C++, SQL, Bash

ML / AI Libraries & Frameworks: PyTorch, Scikit-learn, XGBoost, CatBoost, NumPy, CuPy, Pandas, Optuna, LangChain

Developer Tools: Git, Docker

Work Experience

IT Infrastructure intern

National Bank of Egypt (NBE)

Cairo, Egypt

July 2026

- Designed disaster recovery plans for critical data centers and core banking services.
- Contributed to the IT automation and disaster recovery sector to ensure the resilience of enterprise infrastructure.

AI/ML Trainee

Elevvo Internship Program

Cairo, Egypt

September 2025

- Developed 5 end-to-end ML pipelines across regression, classification, and time-series forecasting tasks using Python, Scikit-learn, and XGBoost, delivering production-ready models with documented results
- Evaluated and compared diverse machine learning models to identify the optimal architecture for each regression, classification, and forecasting task

Summer Intern

Commercial International Bank (CIB)

Cairo, Egypt

August 2025

- Completed CIB's competitive "The Green Leap" program, analyzing ESG data and digital transformation frameworks across Egyptian corporate case studies
- Conducted quantitative research on corporate governance and cybersecurity risk models, presenting findings to senior banking leaders

Projects

Weather Wizard 3000 | Python, LiteLLM, OpenWeatherMap

github.com/Youssef-Naggar/Weather-Wizard-3000

- Built an AI outfit recommendation assistant that recommends outfits based on weather, closet, and preferences, using vectorless RAG and LiteLLM for SDK-agnostic routing.

Dog Breed Classifier | PyTorch, ConvNeXt-V2

github.com/Youssef-Naggar/Dog-Breed-classifier

- Achieved a **90.5% test accuracy** across 120 breeds by fine-tuning ConvNeXt-V2 built with PyTorch
- Applied MixUp data augmentation and automated bounding-box preprocessing to normalize training data and reduce overfitting across 120 imbalanced dog breed classes

Email Fraud Detection Pipeline | PyTorch, XGBoost, CatBoost, Optuna

github.com/Youssef-Naggar/Email-Fraud-Detector

- Engineered an end-to-end fraud pipeline across 447K Enron emails resolving extreme **192:1 imbalance**, training a **PyTorch Autoencoder** to compress 384-d MiniLM embeddings into 64-d latents and derive unsupervised anomaly reconstruction signals
- Architected a **Tri-Stream Stacking Ensemble** (GPU XGBoost, CatBoost with native text dictionaries, LightGBM on TF-IDF n-grams) with a Level-2 **ElasticNet** logit meta-learner, achieving an **F2-score of 0.884** via dynamic threshold tuning

MicroGrad - Custom Neural Network Engine | Python, CuPy

github.com/Youssef-Naggar/NN-from-scratch

- Implemented a custom neural network engine from scratch in Python using vectorized backpropagation, achieving **84% test accuracy** - matching PyTorch baseline performance on equivalent architecture
- Accelerated computational throughput using CuPy to execute matrix operations directly on the GPU, reaching industrial-grade performance

UltraTube - YouTube Media Downloader | Python, Textual (TUI), yt-dlp, Pytest

github.com/Youssef-Naggar/ultratube

- Developed a multi-tab asynchronous Terminal User Interface (TUI) video downloader, featuring parallel downloads, live progress tracking, metadata processing (merging audio tracks, thumbnails, and subtitle tracks) and batch downloading for multiple URLs

Certifications

- **Building RAG Agents with LLMs** - Nvidia DLI, 2026
- **CS50's Intro to Databases with SQL** - Harvard/edX, 2024
- **IBM Data Science Professional Cert.** - IBM/Coursera, 2025
- **Introduction to MCP & Advanced Topics** - Anthropic, 2026
- **Math for ML Specialization** - Imperial College, 2025